

LLM-ERP Product Overview (English) — v3.0 Conversational ERP

One-page snapshot → Full buying guide for decision makers Audience: Owners / Procurement managers / IT directors / System integrators / Consultants

⚡ v3.0 Strategic Pivot Notice (2026-05-15) Pitch updated: "Conversational ERP for 50-100 person factories — open Chrome, type one sentence to Chat, do CRUD on all documents at 1/10 the cost of SAP."

Deprecated sections (moved to Phase 7 pending customer feedback):

- "Outsource LINE Coordination" chapter
- Mobile App / LINE Bot marketing content



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1. One-Page Snapshot

Item	Detail
Product	LLM-ERP (AI-Native ERP for SMB Manufacturers)
Positioning	The LINE-Native ERP for 50-100 person factories
Core Promise	The owner runs the factory from LINE; live in 2 weeks, productive in 2 hours
Three Weapons	① Natural language ops ② Mobile-first ③ Outsource LINE collaboration
Tech Foundation	FastAPI + PostgreSQL/SQLite + Multi-Agent LLM + React/Expo
Deployment	One-click Docker, self-host / cloud / MESH multi-factory
Language	Traditional Chinese + English (full i18n)
Devices	Desktop browser + iOS app + Android app + LINE Bot
Onboarding Time	< 2 weeks (vs SAP 6-18 months)
Training / employee	< 2 hours (vs traditional 1-3 months)
Typical Price	NT\$ 300K-500K / year (vs SAP NT\$ 2M+)

In one sentence: We collapse "6-month rollout, 3-month training" pain into "2 weeks live, 2 hours productive" by letting **AI translate the system for users**.

2. Product Positioning

2.1 Ideal Customer Profile (ICP)

Dimension	Range
Headcount	50-100 people (often 10-20 in-house + outsource chain)
Industry	Metal machining / Plastic injection / PCB / Food / Textile dyeing
Revenue	NT\$ 100M-1B / year
Digital maturity	Low-to-mid (still on Excel + LINE groups)
IT budget	NT\$ 300K-1M / year
Owner role	Reads numbers personally, no dedicated IT department

2.2 Five Typical Users (Persona)

Role	Daily Scene	Primary Device
Owner (Mr. Wang)	06:30 asks LINE "today's status" → gets one-page dashboard	LINE
Salesperson (Chen)	In customer's office, looks up M6 stock on phone, quotes in 3s	Phone
Plant Manager (Lin)	On factory floor, gets push notification "WO-2026-100 stuck"	Phone + LINE
Buyer/Warehouse (Ling)	Scans QR on incoming carton, inventory auto-increments	Phone (scan)
Outsource Master (Wu)	LINE receives drawing, snaps photo to report progress, no registration	LINE

2.3 Pains We Solve

Customer Voice	Our Solution
"ERP is too expensive, we can't afford SAP"	Open-source core + NT\$ 300K/year subscription
"6-month rollout? Who can endure that?"	Docker one-click + natural language → live in 2 weeks
"Training employees is hell"	LINE + AI chat interface → no system to learn
"Inventory out of sync, sales quote wrong stock"	Live DB + SSE push + 3-second mobile query
"Outsource shops can't see our progress"	LINE QR + Bot reporting → no registration needed
"Boss wants live numbers, not month-end reports"	LINE Bot real-time dashboard
"We only notice missing parts when it's too late"	EventBus proactive alerts

3. 12 Core Modules

Each domain has a complete RESTful API + desktop UI + mobile UI + natural-language interface.

#	Domain	Key Features	Primary Users
1	Inventory	Parts / BOM / Transactions / Safety stock alerts / Multi-warehouse	Warehouse + Sales
2	Purchase	Supplier tiering (T1-T3) / PO / Receiving / 3-way match / Auto AP	Buyer
3	Sales	Customer / Quote / SO / Shipment / Auto AR	Sales + Owner
4	Production	Work orders / BOM explode / Dispatch / Reporting / OEE	Plant Mgr + Operator
5	Quality	Inspection / Non-conformance / CAPA / Customer complaints	QC + QA
6	Warehouse	Multi-warehouse / Barcode count / Transfer / Auto replenishment	Warehouse
7	MPS/MRP	Master schedule / Material req planning / Simplified for SMB	Planner + Plant Mgr
8	Accounting	AR / AP / Month-end close / Auto journals	Finance
9	CRM	Lead → Opportunity → Customer funnel + Visit log	Sales
10	HR	Employee / Department / Role / Training	HR + Manager
11	AI Governance	DecisionLog (auditable AI calls) / Model switching	IT + Audit
12	Permission	RBAC 5 layers / 109 perms / 11 roles / Row-Level filter	IT + Owner

3.1 Cross-Domain Intelligence

- **Multi-Agent AI:** 10 specialist agents (Inventory / Sales / Production / Quality / Purchase / Planning / Warehouse / Accounting / CRM / General) + 26 tools
- **EventBus:** 16 Constraint Rules auto-fire across domains (e.g. "PO approved" → notify supplier + warehouse)
- **MESH Multi-Factory:** Cross-factory inventory aggregation / VMI (raw data stays in each factory)
- **SSE Live Push:** Inventory changes, WO status, AI alerts reach users in sub-second

4. Key Differentiators

4.1 Natural Language Operation (Killer Feature #1)

```
Owner types: "Today's factory status"
↓
IntentClassifier → ProductionAgent
↓
Tool call: query_today_summary
↓
Reply: 12 WOs today (8 in-progress), 3 items below safety stock,
       5 shipments on track 95%
```

Contrast: Traditional ERP requires 7 menu clicks + 3 reports for the same info.

4.2 Mobile-First Design

Principle	Realization
Mobile-first	5 tabs, big buttons, minimal typing
3-tap rule	Any function ≤ 3 taps
Offline degrade	AsyncStorage caches token + basic data
Large fonts	Readable outdoor by owner / plant manager
One-hand reach	Key actions in thumb zone

4.3 Outsource LINE Collaboration (Industry First)

```
HQ → Via LINE Bot sends drawing + process + one-time QR token
↓
Outsource master (no registration in any system) → opens LINE → scans QR
↓
Reports "80% complete" + photo
↓
HQ work order auto-updates / EventBus triggers QC notification
```

Industry pain: Traditional ERPs require outsource shops to register / learn new software. Masters refuse. **Our solution:** Outsource shops **don't register** at all. If they use LINE, they can use this.

4.4 MESH Multi-Factory Data Sovereignty

HQ — WireGuard VPN —

├ Main Factory (A) :8001

├ Plating Outsource (B) :8002

├ Inspection Outsource (C) :8003

└ Nth Outsource

HQ asks "Total M6 across all factories?" →

• HQ parallel-fans to N factories /api/factory/mesh/query

• Each factory returns only its own aggregated sum (no raw rows)

• HQ totals: 3000 + 1500 + 800 = 5300

★ HQ NEVER sees "which batch, who bought it"

★ Integration test 5/5 PASS – not a stub, actually works

5. Technical Spec

5.1 System Requirements

Resource	Min	Recommended
CPU	2 core	4 core
RAM	4 GB	8 GB
Disk	20 GB SSD	100 GB SSD
OS	Win 10 / Mac / Linux	Linux Server
Docker	20.10+	24+
Network	100 Mbps LAN	1 Gbps
Public access	HTTPS (LINE Bot only)	Cloudflare Tunnel free tier

5.2 Backend Stack

Layer	Tech
Web	FastAPI (Python 3.12+) async
ORM	SQLAlchemy 2.0 async
Database	PostgreSQL 15+ / SQLite (sufficient for SMB)
Migration	Alembic
Auth	JWT (HS256) + RBAC 5-layer
Real-time	Server-Sent Events (SSE)
LLM Adapter	Claude / DeepSeek / OpenAI / Ollama (pluggable)
Container	Docker Compose

5.3 Frontend Stack

Surface	Tech
Desktop UI	React 18 + Vite + Tailwind + Zustand + TypeScript strict
Mobile App	Expo SDK 51 + React Native + expo-router
LINE Bot	LINE Messaging API + Webhook (planned)
i18n	Custom lightweight (zh-TW + en)

5.4 Performance Targets

Operation	Target latency	Measured
Health check	< 100 ms	~20 ms
Login	< 500 ms	~200 ms
List query (<1000 rows)	< 1 s	~300 ms
AI chat (simple)	< 5 s	2-5 s (LLM-dependent)
AI chat (multi-step tool)	< 30 s	6-15 s
MESH aggregate (3 factories)	< 3 s	~500 ms
SSE push latency	< 2 s	< 1 s

5.5 Scale Limits (Single Instance)

Dimension	Limit
Concurrent users	100-200
Requests / sec	200-500 RPS
Inventory parts	100,000
Work orders	100,000 / year
Conversation log	1,000,000 entries
MESH nodes	10-20

Beyond this scale, recommend factory-distributed + MESH aggregation.

6. Deployment Options

6.1 Four Modes

Mode A: Single-machine Docker (simplest, 50-person factory)

- └─ HQ backend + Desktop UI + DB all on one box
- └─ One-click install.bat / install.sh

Mode B: Managed cloud (for factories without IT)

- └─ We host on GCP/AWS/Azure
- └─ Customer sees only https://your-company.llm-erp.app
- └─ Monthly fee covers host + backup + upgrade

Mode C: Customer self-host (data sovereignty)

- └─ Customer buys VM / on-prem server
- └─ We provide Docker image + deployment guide
- └─ Customer's IT maintains

Mode D: MESH multi-factory (group / multi-site owners)

- └─ HQ + N Factory Nodes
- └─ WireGuard VPN mesh
- └─ Each factory's data stays on-site
- └─ HQ sees aggregates only

6.2 Install Time

Mode	First install	Customer does
A — Single Docker	10 min	Double-click install.bat
B — Managed cloud	Next day live	Fill form + pay
C — Self-host	0.5-1 day	Remote pairing with our engineer
D — MESH	1-3 days / factory	Install per factory + setup VPN

7. Integration Capabilities

7.1 Outbound Integrations

Target	Method	Purpose
LINE	Messaging API + Bot Webhook	Owner dashboard / outsource reporting
Excel	CSV / xlsx export	Owners prefer Excel
Email	SMTP (SendGrid-compatible)	PO PDF / reports
3rd-party ERP	REST API + Webhook	Upstream customer / group
Customs / Gov	Reserved interface	Export / customs filing

7.2 LLM Providers (Pluggable)

Provider	Best For	Monthly Cost (50p)
Anthropic Claude	International brand orders, strictest	NT\$ 3,000-8,000
OpenAI GPT-4o	Industry standard	NT\$ 2,500-6,000
DeepSeek	Chinese context, best ROI	NT\$ 800-2,000
Ollama local (gemma3 / qwen2.5)	Zero data egress	NT\$ 0 (one-time hardware)

Switching LLM = edit one `.env` line + restart. See [LLM Benchmark Report](#).

8. Security & Compliance

8.1 Security Architecture

Layer	Mechanism
AuthN	JWT (HS256) + bcrypt password hash
AuthZ	RBAC 5-layer (Tenant→User→Role→Permission→Row-Level)
Audit	AuditMiddleware logs all requests + DecisionLog (AI calls)
Transport	HTTPS (nginx / Cloudflare Tunnel)
Multi-factory	WireGuard VPN + factory VLAN
Secrets	<code>.env</code> + HashiCorp Vault recommended (enterprise)
Demo bypass	Active only when JWT_SECRET is default; auto-disables in production

8.2 Data Sovereignty

- MESH multi-factory mode: Raw rows physically stay in each factory's SQLite/PostgreSQL
- HQ can only call `/api/factory/aggregate` for aggregated numbers
- Integration test verified: aggregate response contains **none** of `created_at` / `qty_available` / `items` / `rows`

8.3 Compliance Readiness

Standard	Status
GDPR (right to erasure)	API endpoint reserved, customer enables
ISO 27001	Documentation 70% complete, requires 3rd-party audit
China PIPL	Can deploy to China DC (DeepSeek/Ollama)
HIPAA	Healthcare requires custom encryption

9. Onboarding Process & Timeline

9.1 Standard Rollout (2-Week Model)

Day 1-2	Contract + Requirements interview
Day 3-5	Environment prep (server / accounts / network)
Day 6-7	Docker install + industry-sample data load
Day 8-9	Customer data import (from Excel)
Day 10	Admin training (2 hours)
Day 11-12	Department head training (5 roles × 1 hour)
Day 13	Internal pilot
Day 14	Go-live

9.2 Extended Rollout (Multi-factory MESH, 4-6 weeks)

Additional: per-factory install, WireGuard VPN setup, cross-factory permissions, aggregation tests.

9.3 Training Hours

Role	Hours
Owner / Plant Mgr	1 hour (LINE-based)
Sales / Buyer	2 hours (incl. mobile app)
Warehouse / Operator	3 hours (incl. barcode scan)
IT Admin	1 day (incl. RBAC + deployment)

Compare with industry-average 1-3 months for SAP.

10. Service & Support

10.1 Service Level Agreements

Tier	Response Time	Applicable
Basic	Business day, 8h	Standard subscription
Pro	Business day, 2h	Add-on
24/7	Anytime, 1h	Enterprise

10.2 Support Channels

- Internal IT / integrator
- Monthly online training
- LINE customer support group
- Emergency phone (Pro+)
- Remote TeamViewer assist (Pro+)

10.3 Upgrade & Maintenance

- **Minor:** Monthly, auto-built + customer clicks upgrade
 - **Major:** Quarterly, new features + training session
 - **Data migration:** Auto migration test across major versions
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11. Procurement FAQ

Q1: Why so much cheaper than SAP / Oracle?

Because we don't charge license fees. SAP's NT\$ 2M is mostly "license" + "implementation consultant". We're open-source core + subscription (annual fee). Customers buy subscription + service, not "ownership of software".

Q2: Isn't open-source risky?

No. LLM-ERP is a commercial project; **open-source core** lets customers: ① see the code ② leave anytime without lock-in ③ find someone else to maintain. **Our value is in service**, not in trapping customers.

Q3: Is our data safe?

Four layers of protection:

1. **Architectural:** MESH mode keeps data in each factory
2. **Transport:** HTTPS + WireGuard VPN
3. **Storage:** bcrypt passwords + optional disk encryption
4. **Audit:** AuditMiddleware logs all + DecisionLog

Can choose Ollama local LLM for **zero data egress**.

Q4: Can we customize?

Yes, three ways:

- **Configuration:** change settings via admin UI (free)

- **Extension:** write new domain / agent / tool (consulting fee)
- **Core modification:** open source so you can fork (manage your own fork)

Q5: What if your company goes under?

Open-source core continues to work; data is in customer's hands. Any Python engineer can maintain it (widespread stack). MESH mode means factory data is fully self-owned, even safer.

Q6: Payment terms?

- Annual (standard)
- Quarterly (+10%)
- One-time license (self-host mode only, $\approx 3 \times$ annual)

Q7: Can we try first?

Yes. **Demo Mode 14-day free trial**, no signup:

1. Double-click `install.bat`
2. Click "Demo Mode" button
3. Use sample data to explore everything

Q8: Do you have training materials?

Complete set:

- User Manual (ZH/EN)
- Installation Guide (ZH/EN)
- System Architecture (ZH/EN)
- LLM Benchmark Report (ZH/EN)
- Network Deployment Plan (ZH/EN)
- Video tutorials (planned)

All exportable to PDF (`build_pdfs.bat`).

12. Competitive Comparison

12.1 Three Solutions

Dimension	SAP Business One	Odoo Enterprise	LLM-ERP
License / year	NT\$ 1-2M	NT\$ 300-800K	NT\$ 300-500K
Rollout time	6-18 months	2-4 months	< 2 weeks
Training / employee	1-3 months	2-4 weeks	< 2 hours
Mobile UX	Weak (legacy UI)	Mid (responsive)	Strong (native app)
AI natural language	None	None	Core feature
LINE Bot integration	Custom	Custom	Built-in
Outsource LINE collab	None	None	Industry first
Multi-factory sovereignty	Centralized	Centralized	MESH
Customization	Low (closed)	Mid (open)	High (open + AI)

12.2 Why Not Just Excel?

Scenario	Excel	LLM-ERP
Owner asks "today's status"	Flip 5 sheets	One sentence reply
Sales checks stock	Open front desk PC	3-second mobile
Real-time sync	Impossible	SSE push
Outsource collaboration	Chaotic LINE group	QR + auto-update
Audit trail	None	AuditMiddleware full log

13. How to Buy

13.1 Purchase Process

Step 1

Download and install → Demo Mode trial (14 days)

Step 2

Fill requirements form (industry / size / customization)

Step 3

Our consultant 30-min requirements interview

Step 4

Quote (annual fee + one-time implementation)

Step 5

Sign + 50% prepayment

Step 6

Day 1 onboarding starts

Step 7

Day 14 go-live + final payment

13.2 Contact

Need	Contact
Trial issues	LINE @llmerp
Sales inquiry	sales@llm-erp.example
Tech support	support@llm-erp.example
Integrator partnership	partners@llm-erp.example

Related Documents

- [Installation Guide \(English\)](#) / [中文](#)
- [User Manual \(English\)](#) / [中文](#)
- [System Architecture & Topology](#) / [中文](#)
- [LLM Benchmark Report](#) / [中文](#)
- [Network Deployment Plan](#) / [中文](#)
- Chinese version: `PRODUCT_OVERVIEW_ZH.md`

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